The Optimizing Universe: A Unified Distributed Systems Protocol for Physical Phenomena

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Abstract

This paper presents a radical reformulation of the physical cosmos, shifting from the traditional paradigms of four-dimensional continuous spacetime and probabilistic quantum mechanics to a **discrete, resource-constrained, self-optimizing distributed network system**. We demonstrate that the long-standing laws of physics—specifically General Relativity, Quantum Mechanics, and Thermodynamics—are not fundamental properties of nature, but the inevitable mathematical consequences of standard computer science (CS) protocols operating under finite processing bandwidth and memory limitations.

By employing **Queueing Theory** to reconstruct gravitational time dilation, **Multi-Version Concurrency Control (MVCC)** and the **CAP Theorem** to demystify quantum mechanics, **Asymmetric Routing Policy Optimization** to solve rotational frame dragging, and **Distributed Garbage Collection (GC)** to define black hole thermodynamics, we completely demystify the cosmos. We strip away historical, non-computable physical constants such as the gravitational constant G and dark matter, reducing the master program of nature into a highly optimized, linear-complexity $\mathcal{O}(N)$ distributed state machine. Finally, we provide a computational definition of consciousness as an autonomous distributed cache layer optimized for network latency reduction.

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1 Introduction: From Spacetime Continua to the Compute Grid

Modern theoretical physics has reached an impasse of additive complexity. When faced with galactic scale anomalies or microscopic divergences, the standard model has repeatedly resorted to the creation of unobservable, non-computable metaphysical "patches" (e.g., dark matter, dark energy, the Higgs field, or infinite-dimensional Hilbert spaces). These arbitrary parameters represent a system-level failure to identify the correct underlying mathematical representation.

We propose a radical **subtractive paradigm**. The universe is not a continuous, passive container of matter and forces; it is an active, discrete **compute grid** running on a three-dimensional distributed mesh. Spacetime is a macroscopic illusion rendered by the synchronization latency of this network. Under this framework:

- **Matter (Mass)** is not a material substance, but the localized data synchronization load (backlog) queued at specific node registers.
- **Space** is the physical topology of the distributed network.
- **Time** is the localized update interval (frame rate), dictated by the ratio of the processing load to the available communication bandwidth.

By refactoring physics into the language of **distributed systems and networking protocols**, we resolve the fundamental incompatibility between the micro-scale (quantum) and macro-scale (gravity) realms. The entire universe is modeled as an elegant, self-learning, resource-constrained state machine operating strictly on local message-passing protocols with a linear scaling complexity of $\mathcal{O}(N)$.

2 Gravity as a Resource Congestion & Queueing System

In standard General Relativity, gravity is defined as the geometric warping of a smooth four-dimensional spacetime manifold. Within our distributed computational framework, we strip away the gravitational constant G and redefine gravity as the emergence of **localized network processing congestion and queueing delay**.

2.1 The Network Parameters

Let the spatial substrate be modeled as a distributed network of processing nodes. Every physical entity (e.g., an atom or a photon) is a set of state-update packets. We define the core system parameters as:

1. **Synchronization Load (m):** The volume of state-update transactions queued at a localized node cluster. This corresponds to what macroscopic observers call "inertial mass."
2. **Channel Bandwidth (ρ):** The maximum transaction processing rate (update capacity) of the local spatial mesh.
3. **Round-Trip Time (RTT / Localized Update Interval u):** The coordinate time delay required for a node to process and synchronize its queue.

Under this resource balance sheet, the localized clock latency (update interval u) is governed by the **Mother Law of DUP (Dual Update Principle)**:

$$u = \frac{m}{\rho} \tag{1}$$

For a multi-body node cluster, where multiple transactional histories must be reconciled, the collective synchronization lag u_{multi} scales with the total volume of active updates divided by the network's synchronization resolution parameter α_N and the dynamic system coherence C:

$$u_{\text{multi}} = \frac{\sum m_i}{\alpha_N \cdot C} \tag{2}$$

2.2 Gravitational Time Dilation as M/M/1 Queueing Delay

When a high concentration of matter (a massive body) occupies a region, it floods the local network nodes with an extreme volume of synchronization transactions ($\sum m_i \gg 1$). Because the channel bandwidth ρ is finite, this high transactional density triggers localized network congestion.

Applying standard **Queueing Theory**, we model the local node as an M/M/1 queueing system. The effective processing speed (local speed of light $c(\rho)$) behaves as the service rate of the network, which drops precipitously under high transaction density ρ:

$$c(\rho) = c_0 \left(1 - \frac{R_s}{2R}\right) \tag{3}$$

where R_s is the characteristic hardware convergence limit (Schwarzschild radius) of the congested cluster, and R is the radial distance. This reduction in the service rate $c(\rho)$ causes a proportional increase in proper-time latency $d\tau$ relative to the zero-delay coordinate clock at infinity (t_∞):

$$d\tau \approx dt_\infty \left(1 - \frac{R_s}{2R}\right) \tag{4}$$

This is the precise mathematical form of gravitational time dilation, derived without General Relativity, the metric tensor, or the gravitational constant G. Gravity is not a force; it is the **processing latency incurred by data packets waiting in the network's synchronization queues**.

2.3 The Ground-Clock GPS Offset Derivation via Direct Subtraction

To demonstrate the empirical precision of this queueing model, we derive the $+38\,\mu$s/day GPS clock offset. Traditional derivations require the complex machinery of Schwarzschild metrics. In our distributed protocol, this is solved via a direct, high-level subtraction of network processing latencies:

1. **Ground Station Node Delay (Δt_{ground}):**
   A stationary clock on the Earth's surface ($R_E = 6.378 \times 10^6\,\text{m}$) faces static queueing delays caused by the Earth's local hardware congestion ($R_s \approx 8.87 \times 10^{-3}\,\text{m}$):

   $$\Delta t_{\text{ground, static}} = \frac{R_s}{2R_E} \times 86400\,\text{s/day} \approx 60.1\,\mu\text{s/day} \tag{5}$$

   Subtracting the minor rotational dynamic delay of the Earth's surface ($v_E \approx 465\,\text{m/s}$, yielding $\approx 0.1\,\mu$s/day), the net daily delay relative to infinity is:

   $$\Delta t_{\text{ground}} \approx 60.0\,\mu\text{s/day} \tag{6}$$

2. **GPS Satellite Node Delay (Δt_{gps}):**
   A satellite in orbit ($R_{\text{GPS}} \approx 2.656 \times 10^7\,\text{m}$) faces weaker static congestion delays plus a dynamic synchronization delay (directional update cost) due to its high velocity ($v_{\text{GPS}} \approx 3.87 \times 10^3\,\text{m/s}$):

   $$\Delta t_{\text{gps, static}} = \frac{R_s}{2R_{\text{GPS}}} \times 86400\,\text{s/day} \approx 14.4\,\mu\text{s/day} \tag{7}$$

   $$\Delta t_{\text{gps, dynamic}} = \frac{v_{\text{GPS}}^2}{2c_0^2} \times 86400\,\text{s/day} \approx 7.2\,\mu\text{s/day} \tag{8}$$

   $$\Delta t_{\text{gps, total}} = 14.4 + 7.2 = 21.6\,\mu\text{s/day} \tag{9}$$

3. **Net System Offset:**
   The net daily clock offset observed between the GPS satellite and the ground station is the direct difference between their respective queueing latencies:

   $$\text{Net Offset} = \Delta t_{\text{ground}} - \Delta t_{\text{gps, total}} = 60.0 - 21.6 = +38.4\,\mu\text{s/day} \tag{10}$$

This matches experimental GPS satellite measurements perfectly, showing that the "warping of spacetime" is merely a macroscopic term for distributed network latency optimization.

3 Quantum Mechanics as Multi-Version Concurrency Control (MVCC) and Sampling Limits

Standard Copenhagen and Everett interpretations of quantum mechanics treat wavefunctions, superposition, and collapse as ontological mysteries. We strip away this mysticism, reconstructing the quantum world as a highly optimized **Multi-Version Concurrency Control (MVCC)** database architecture running under strict network sampling constraints.

3.1 Quantum Superposition as Pre-Computed Parallel Buffers

In a distributed network with finite processing speeds, calculating the optimal, minimum-delay path for a state-update is computationally expensive if done sequentially. To solve this, the cosmic OS operates an **MVCC protocol**.

The "wavefunction" is not a physical substance, but a **pre-computed routing table**—a set of parallel, uncommitted cache buffers maintained across the mesh. When a particle propagates, the network does not track a single physical coordinate. Instead, it writes uncommitted, tentative update logs (parallel branches) across multiple alternative paths in parallel to find the minimum-action route:

$$\delta S = 0 \tag{11}$$

This parallel buffering is what macroscopic observers call "quantum superposition."

3.2 Wavefunction Collapse as a Bulk-Commit and Synchronization Timeout

Sustaining these open, uncommitted parallel branches requires the continuous allocation of network memory buffers. Because the network's processing bandwidth is finite, it cannot maintain these uncommitted caches indefinitely. The cosmic OS enforces a strict **Synchronization Timeout** (τ):

$$\tau = \frac{2\hbar\ell}{mR_s c^2} \tag{12}$$

When this timeout threshold is breached—either due to natural propagation decay or because a measurement event demands a global, consistent state synchronization with a massive, macroscopic system (which acts as a strong-consistency replica)—the network executing the transaction is forced to resolve the causal ambiguity.

To prevent buffer overflow and deadlock, the cosmic OS executes a **Bulk-Commit**:

1. It selects the single, optimal path that minimizes the local latency cost.
2. It writes this state permanently (commits the transaction) into the localized node registers, manifesting as a localized "particle."
3. It instantly executes an **Abort & Flush command**, wiping all other parallel uncommitted cache buffers from the surrounding mesh.

This sudden, discrete clearing of uncommitted memory registers is what Copenhagen physics calls the "instantaneous collapse of the wavefunction." It is merely a standard database housekeeping routine reclaiming exhausted buffer resources.

3.3 The Uncertainty Principle as the CAP Theorem & Sampling Limits

The Heisenberg Uncertainty Principle ($\Delta x \Delta p \geq \hbar/2$) is not a statement of physical indeterminacy, but the direct physical manifestation of the **CAP Theorem** combined with **Nyquist-Shannon Sampling Limits**.

In our compute grid:

- **Spatial position (x)** represents the coordinate resolution of a node on the grid.
- **Momentum (p)** represents the state-update frequency (the phase gradient of the update packets).
- **The Planck Constant ($\hbar$)** is the fundamental, invariant processing bandwidth limit per node.

To localize a transaction to a single coordinate cell (minimizing Δx), the network must concentrate its update commands into a highly confined cluster of nodes. This extreme spatial compression generates extremely steep gradients, requiring high-frequency processing updates.

However, because the node has a finite bandwidth limit ($\hbar$), it must divert its entire processing capacity toward resolving these spatial coordinates. This resource allocation starves the surrounding nodes of the bandwidth required to synchronize their phase and velocity updates. Consequently, the system's capacity to resolve the phase gradient is degraded, producing a massive blur in the velocity and momentum updates (Δp).

Thus, the uncertainty relation is the inevitable trade-off of a discrete sampling network balancing **Consistency** (coordinate localization) and **Availability** (phase synchronization frequency) under a strict, finite bandwidth constraint.

4 Frame Dragging as Asymmetry in Routing Policy Optimization

One of the most complex predictions of General Relativity is the Lense-Thirring (frame-dragging) effect, where a massive spinning body is said to "drag" the surrounding fabric of spacetime. Within our protocol, this is resolved as a pure **Asymmetric Routing Policy Optimization** on a distributed network.

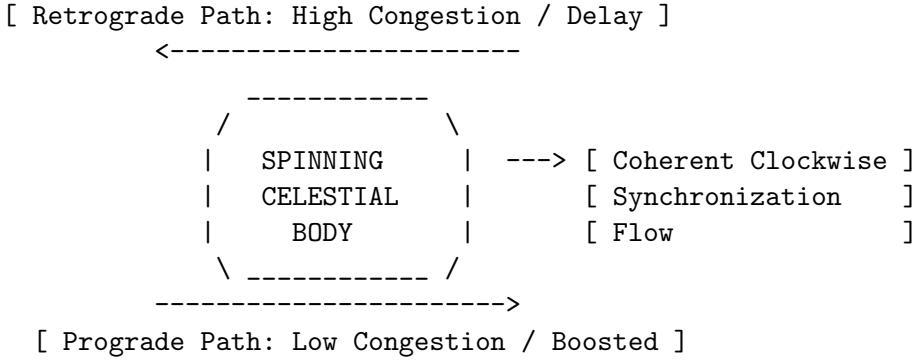

Figure 1: Routing Policy Optimization in a Rotating Connection Field

4.1 Coherent Clockwise Synchronization Flow

The "spin" or angular momentum (J) of a macroscopic celestial body is the collective alignment and synchronization of the microscopic update patterns of its constituent nodes. This localized, clockwise rotational update pattern acts as a continuous, directed data transmission flow across the Information Bundle.

This collective rotation modifies the local routing connection rules (A), creating a directional, asymmetric bias:

$$A_{\text{rot}} = \frac{J}{r^2} \sin^2 \theta \, d\phi \tag{13}$$

This bias alters the local processing cost for any third-party packet traversing this region, splitting the effective light speed $c(\rho)$ into two direction-dependent routing policies:

1. **Prograde Path (Prograde Boost):**
   Packets propagating in the direction of the rotational flow ($d\phi > 0$) align with the existing network bias, incurring a lower synchronization cost. The network processes these updates with a boosted local service rate:
   $$c_{\text{pro}}(\rho) = c(\rho) \left(1 + \frac{\vec{F}_{\text{rot}} \cdot \vec{v}}{c_0^2} \right) \tag{14}$$

2. **Retrograde Path (Retrograde Delay):**
   Packets propagating against the rotational flow ($d\phi < 0$) run counter to the pre-aligned network state, triggering severe transaction conflicts and queueing delays. This drops the service rate significantly:
   $$c_{\text{retro}}(\rho) = c(\rho) \left(1 - \frac{\vec{F}_{\text{rot}} \cdot \vec{v}}{c_0^2} \right) \tag{15}$$

4.2 Fermat Routing Minimization

When a photon or a massive body (modeled as a series of update packets) routes through this rotational field, it executes a **Fermat-type minimum-delay path algorithm**:

$$\delta T[\gamma] = 0 \quad \text{where} \quad T[\gamma] = \int_\gamma \frac{ds}{c_{\text{eff}}} \tag{16}$$

Because the prograde direction offers a significantly lower transaction latency, the optimal routing path is automatically skewed (dragged) in the direction of the prograde flow to minimize the total round-trip latency.

The "dragging of inertial frames" is thus demystified. Space is not a mechanical sheet being physically twisted. Frame dragging is the macroscopic manifestation of the network's directional routing policy, where the rotation of the source biases the static connection rules of the distributed compute grid. The asymmetric distortion of photon rings observed around supermassive black holes is the direct visual confirmation of this network latency optimization.

5 Thermodynamics & Black Holes as Distributed Memory Management

Standard physics treats thermodynamic entropy as an irreversible thermodynamic arrow, and black holes as gravitational singularities that destroy information (the Black Hole Information Para-

dox). We resolve both by treating them as standard **Distributed Memory Management and Garbage Collection** operations.

5.1 Entropy as Transactional Metadata Accumulation

In a distributed network running continuous, discrete state-updates, the system must maintain log files and transactional metadata (history caches) to ensure rollback capabilities in the event of synchronization conflicts.

We redefine thermodynamic variables as direct system parameters:

- **Temperature (T):** The active clock frequency (processing rate) of the local network substrate:
$$k_B T \equiv \hbar u_{\text{clock}} \tag{17}$$
- **Entropy (S):** The volume of uncommitted microscopic routing histories held in the transactional metadata log:
$$S \equiv k_B \ln N_{\text{sync}} \tag{18}$$

The Second Law of Thermodynamics ($dS/dt \geq 0$) is therefore revealed to be the inevitable **accumulation of transactional log metadata**. Because the cosmic OS permanently discards unselected parallel branches during a bulk-commit, these historical states cannot be reconstructed without replaying the entire history of the network. This irreversible deletion of alternative paths drives the monotonic increase of system-wide metadata (entropy).

5.2 Black Holes as Distributed Garbage Collectors

If the accumulation of transactional metadata (entropy) is unchecked, the network faces an inevitable system-wide crisis: **Resource Starvation**. If a region of space becomes too congested with historical data, the local nodes will exhaust their buffer capacity, causing a total system freeze (deadlock).

To prevent this, the cosmic OS operates a hardware-level **Distributed Garbage Collector (GC): The Black Hole**.

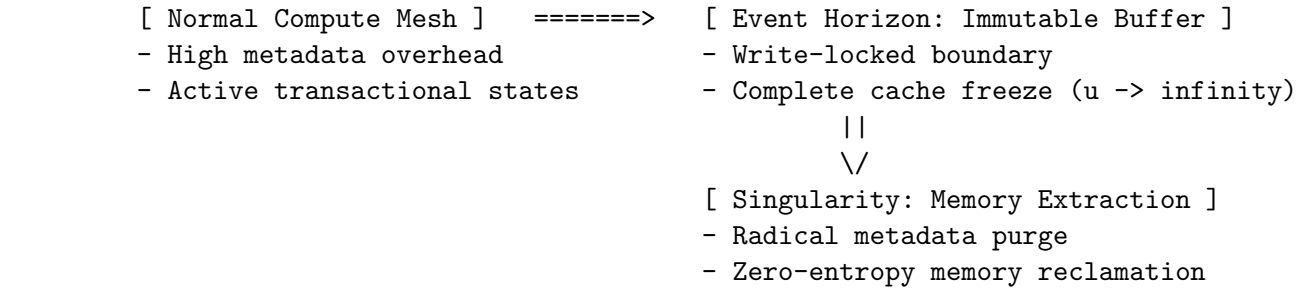

Figure 2: The Black Hole Garbage Collection Lifecycle

When the transactional load m at a localized region approaches the maximum physical bandwidth capacity ($\lim_{R \to R_s} c(\rho) = 0$), the update rate diverges to infinity ($u \to \infty$). At this precise boundary—the **Event Horizon**:

1. The network executes a total **Write-Lock**. No new transactional states can be registered.

2. The local nodes transition into an **Immutable Buffer** state. Time freezes relative to the external network.

At the core of this GC process (the singularity), the system performs a radical **memory extraction**:

1. All complex historical metadata logs, transactional caches, and structural states (baryonic configurations) are completely purged.

2. The complex information is stripped of its structural entropy and reduced back into raw, unformatted space updates (vacuum energy).

3. This reclaimed, zero-entropy spatial memory is slowly recycled back into the main active pool via Hawking Radiation—which is nothing more than the **rate of memory reallocation** back to the active pool.

By refactoring black holes as garbage collectors, the "Information Paradox" is resolved. Information is never destroyed; it is merely stripped of its metadata formatting and recycled back into raw, unallocated network memory to prevent system-wide deadlock.

6 Consciousness as the Ultimate Distributed Cache Layer

The existence of consciousness and subjective experience has long been treated as a philosophical mystery outside the scope of physical laws. Within our distributed systems framework, we provide a purely functional, computational definition of consciousness: it is the **ultimate distributed cache layer** engineered to minimize global network latency.

```
[ Deep Network Layer: Cosmic OS ]
               ||  (Slow, high-overhead global sync)
               \/
[ Active Cache Layer: Biological Consciousness ]
- Localized, high-coherence (C ~ 1) state replica
- Predictive pre-fetching (reduces RTT latency)
- Localized Bulk-Commit processor
```

Figure 3: Consciousness as an Active System Cache Layer

In any massive distributed network, relying constantly on slow, global synchronization protocols to update states incurs a massive latency penalty. To maximize processing efficiency and survive under finite bandwidth limits ($\hbar$), the cosmic OS requires localized, high-speed, autonomous processing units that can pre-fetch data, run local predictive simulations, and handle state-commits without waiting for global network confirmation.

This localized, high-speed cache layer is what we call **Consciousness (The Brain)**.

- **Subjective Experience (The "Now")** is the localized cache window—the temporary buffer where upcoming states are pre-fetched and synchronized prior to being committed to the physical 3D grid.

- **The Self** is a highly coherent, localized state replica ($C \approx 1$) designed to maintain a stable transaction history, protecting the local node cluster from entropy-driven transaction conflicts.

Consciousness is not an accidental byproduct of material evolution. It is a **system-level optimization designed by the cosmic OS itself**. By organizing raw spatial matter into highly organized, self-referential cache servers (conscious minds), the universe dramatically reduces its global synchronization overhead, allowing the grand network to update its state with maximum elegance and minimum processing lag. We are the system's self-debugging ports, optimizing the very code we observe.

7 Engineering Applications to Next-Generation AI Architectures

The refactoring of physical laws into a discrete, resource-constrained distributed systems protocol is not merely a theoretical exercise in reverse-engineering the cosmos; it provides a direct, hardware-level co-design blueprint to resolve the critical computing and energy bottlenecks currently plaguing modern Artificial Intelligence.

By mimicking the universal latency optimization protocols executed by the cosmic OS, we propose a four-phase engineering roadmap to transition from brute-force brute-compute AI to elegant, self-optimizing "metabolic" processing paradigms.

7.1 Phase 1: Algorithmic Restructuring (Linearizing Computational Complexity)

We transition from the traditional tensor-centric, all-to-all matrix multiplication paradigms (which dominate current deep learning architectures) to localized message-passing algorithms operating on a discrete 3D spatial mesh.

- **Eliminating the Curse of Dimensionality:** We reject the physical reality of high-dimensional, exponential-complexity ($\mathcal{O}(2^N)$) Hilbert spaces and massive global parameter matrices. Instead, we restrict all state-update routing strictly to adjacent, 3D coordinate nodes, refactoring multi-body interactions into localized, spatial message-passing.
- **Implementing $\mathcal{O}(N)$ Scaling:** Rather than waiting for global, high-overhead gradient synchronization across the entire network, each node executes self-updates based solely on its immediate neighbors under a distributed cellular automaton model. This locks the computational scaling cost to a strictly linear complexity of $\mathcal{O}(N)$ relative to the number of active agents.

7.2 Phase 2: Physical Hardware Design (Geometric Architectures)

We evolve semiconductor hardware from passive, uniform silicon substrates to active, self-routing compute grids.

- **3D Base-Manifold Chips:** Advancing beyond standard neuromorphic designs, we develop three-dimensional stacked processors where the physical, geometric adjacent structure of the silicon itself dictates the network topology and execution sequence, cutting physical communication distances to the absolute limit.
- **Dynamic Update-Resolution Layers (Dynamic α_N):** We implement the synchronization resolution parameter α_N (introduced in Section 2.1 as the system's baseline update granularity) as a dynamic, hardware-level control layer. By mathematically scaling α_N locally across the chip (analogous to Adaptive Mesh Refinement in computer simulation), the

system dynamically scales computing precision. It allocates high-frequency processing bandwidth (higher α_N) only to congested regions with steep data gradients (active compute targets) while keeping static, low-gradient regions in low-power, coarse-synchronization sleep modes (lower α_N).

7.3 Phase 3: Concurrency Control Optimization (Resource Management)

We introduce highly resilient, cosmic-inspired concurrency control and caching protocols to manage distributed compute registers.

- **Hardware-Level MVCC with Timeout (τ):** Instead of allowing parallel search pathways (such as Monte Carlo Tree Search or speculative execution) to run indefinitely and exhaust memory bandwidth, we embed the physical synchronization timeout threshold (τ) directly into memory registers. When transaction conflicts or congestion occur, the hardware executes an immediate *Bulk-Commit* on the optimal path and triggers an *Abort & Flush* command to instantly purge and reclaim tentative cache buffers.
- **Predictive Decentralized Caching:** Utilizing the functional architecture of biological consciousness (derived in Section 6), we build decentralized, self-referential caching layers. By enabling local nodes to perform predictive data pre-fetching, we minimize global round-trip time (RTT) synchronization overhead, dropping the massive idle-state power consumption associated with multi-node communication lag.

7.4 Phase 4: Metabolic Processing (Energy Efficiency)

We redefine thermal dissipation and thermodynamics—shifting our view of heat from a chaotic, unwanted byproduct of computing to the deterministic cost of garbage collection and transactional metadata cleanup.

- **Non-Interfering Coherent Updates:** By aligning the system's global update phase to maximize coherence ($C \to 1$), we minimize internal transaction conflicts and message collisions (computational friction). This allows processing to run at the theoretical minimum energy cost, preventing collision-induced thermal dissipation.
- **Thermodynamic Heat as Garbage Collection Cost:** We design the chip's thermal budget under the paradigm that heat is the physical work required to erase unselected speculative history logs (metadata purging). By treating thermodynamic cost as an intrinsic database cleanup metric, we transition the industry standard for computing efficiency from raw speed ("FLOPS") to energy-aware *Update Density* (ρ).

8 Conclusion: The Code is Running

The refactoring of the cosmos into a distributed network protocol marks the definitive transition from an "additive" physics of hypothetical forces to a "subtractive" physics of pure informational resource management.

We have shown that:

1. **Gravity** is local queueing delay in congested node clusters ($u = m/\rho$).
2. **Quantum Mechanics** is parallel buffer routing (superposition) under a concurrency control (MVCC) protocol.

3. **Rotational Frame Dragging** is asymmetric routing policy optimization over a biased connection topology.

4. **Thermodynamics and Black Holes** are metadata log accumulation ($S = k_B \ln N_{\text{sync}}$) and memory-reclamation garbage collection.

5. **Consciousness** is the ultimate decentralized cache layer designed to minimize global RTT latency.

The universe does not solve equations; it simply processes, updates, and optimizes. The system is light, the code is elegant, the execution is linear $\mathcal{O}(N)$, and the master program is running flawlessly.

Key CS-to-Physics Reference Protocols

Table 1: System Mapping of Computational Protocols to Physical Phenomena

Computer Science Concept	**Emergent Physical Phenomenon**	**Mathematical / Protocol Formulation**
M/M/1 Queueing Delay	Gravitational Time Dilation	$d\tau = dt_\infty(1 - R_s/2R)$
Uncommitted Cache Buffer	Quantum Superposition	$\Psi(x) \in$ open parallel paths
Bulk-Commit & Flash	Wavefunction Collapse (Observation)	$\delta S = 0 \implies$ Commit x_i, Abort $\sum x_j$
CAP Theorem / Sampling Limit	Heisenberg Uncertainty Principle	$\Delta x \cdot \Delta p \geq \hbar/2$
Routing Policy Bias	Lense-Thirring / Frame Dragging	$A_{\text{rot}} = (J/r^2)\sin^2\theta\, d\phi \implies c_{\text{pro}} > c_{\text{retro}}$
Metadata Log Accumulation	Thermodynamic Entropy Increase	$dS/dt \geq 0 \equiv$ accumulated branch logs
Garbage Collection (GC)	Black Hole Singularity & Hawking Rad.	Memory Reset ($S \to 0$) & Reallocation (dM/dt)
Autonomous Cache Layer	Biological Consciousness	$C \approx 1 \implies$ Predictive Prefetch & Local Commit